Anexo 6: Reporte de Ajuste de Red Geodésica GNSS

Contiene los resultados del ajuste 3D por mínimos cuadrados de la red geodésica conformada por las estaciones de control y los puntos de proyecto, incluyendo las pruebas estadísticas globales (Chi-cuadrado y F-Test), el análisis de fiabilidad externa y las coordenadas definitivas en el sistema de referencia oficial MAGNA-SIRGAS (Origen Nacional).

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- when it has to be right

Network Adjustment Report

Report created: 06/10/2026 19:04:15

Project Details

General		Customer Details		Master Coordinate System	
Project Name:	Daniel Cortes	Customer Name:	-	Coordinate System Name:	MAGNA-SIRGAS_Origen-Nacional
Owner:	-	Contact Person:	-	Transformation Type:	Classical 3D
Lead Surveyor:	-	Number:	-	Residual Distribution:	None
Date Created:	06/10/2026 18:18:57	Email:	-	Ellipsoid:	GRS 1980
Last Accessed:	06/10/2026 18:18:57	Skype:	-	Projection Type:	Transverse Mercator
Application Software:	Infinity 3.0	Website:	-	Geoid Model:	Geocol2004
				CSCS Model:	-

Path:	C:\Users\usuario\Documents\Leica Geosystems\Infinity\Projects\Daniel Cortes\Daniel Cortes.iprj
Size:	57,2 MB
Comments:	-

Name:	10/06/2026 19:03:47
Date/Time:	06/10/2026 19:03:47

Processing kernel:	MOVE3 4.4.1

www.MOVE3.com

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Adjustment Settings

General	
Controls:	Constrained
Dimension:	3D
Coordinate system:	WGS84
Height mode:	Ellipsoidal
Confidence Level for Heights - 1D:	68.3%
Confidence Level for Error Ellipses - 2D:	39.4%
Max. number of Iterations:	3
Iteration criteria:	0,0001 m

Standard Deviation Settings

Source of Standard Deviation Settings - TPS:	**Individual**
Source of Standard Deviation Settings - GNSS:	**Individual**
Source of Standard Deviation Settings - Level:	**Individual**
Source of Centring/Height Errors - TPS:	**Individual**
Source of Centring/Height Errors - GNSS:	**Individual**

Testing Criteria

Level of Significance (α, multi-dimensional):	0,16

Level of Significance (α0, one-dimensional):	5.0%
Power of Test (1-β):	80%
σ a-priori:	1,0
σ a-priori (GNSS):	25,0

Advanced

Use reduced observations:	No

Adjustment Summary

Adjustment

Type:	Constrained
Dimension:	3D
Number of Iterations:	1
Max. coordinate correction in last iteration:	0,0000 m

Stations

Number of (partly) known stations:	2
Number of unknown stations:	2
Total:	**4**

Observations

GNSS coordinate differences:	12 (4 baselines)
Known coordinates:	4
Total:	**16**

Unknowns

Coordinates:	12
Total:	**12**

Degrees of freedom:	**4**
Optimisation criterion v'Pv:	**5,60486**
σ a-posteriori:	**1,184**

Testing

Critical value W-Test:	1,96
Critical value T-Test (2 dimensional):	2,42
Critical value T-Test (3 dimensional):	1,89
Critical value F-Test:	1,63
F-Test:	1,40

Chi Square Test (95,0%)	
Critical value lower bound:	0,12
Critical value upper bound:	2,79
Chi Square Test:	1,40

Input data

Approximate Coordinates

Station	WGS84 Latitude	WGS84 Longitude	Ellip. Height [m]	Description
BUEN	3° 52' 55,29063" N	77° 00' 37,51018" W	57,7645	Control Fixed 1D
CALI	3° 22' 32,84048" N	76° 31' 57,23035" W	1.027,4846	Control Fixed 3D
GPS-02	3° 29' 03,95343" N	76° 11' 52,87418" W	1.535,5400	Averaged
gps1	3° 29' 05,31530" N	76° 11' 53,04963" W	1.536,0807	Averaged

Observations

From Station	Target Point	ΔX [m]	ΔY [m]	ΔZ [m]	SD ΔX [m]	SD ΔY [m]	SD ΔZ [m]
BUEN	GPS-02	88.800,6581	16.728,8950	-43.785,3228	0,0117	0,0263	0,0094
CALI	GPS-02	36.084,5461	8.970,8317	12.024,7957	0,0087	0,0192	0,0069
CALI	gps1	36.078,8082	8.971,4850	12.066,5928	0,0098	0,0181	0,0065
BUEN	gps1	88.794,9421	16.729,3712	-43.743,5067	0,0392	0,0866	0,0281

Adjustment Results

Adjusted Coordinates

Station	WGS84 Latitude	WGS84 Longitude	Ellip. Height [m]	Corr WGS84 Lat [m]	Corr WGS84 Long [m]	Corr Height [m]
BUEN	3° 52' 55,29640" N	77° 00' 37,51286" W	57,7645	0,1773	-0,0826	0,0000
CALI	3° 22' 32,84048" N	76° 31' 57,23035" W	1.027,4846	0,0000	0,0000	0,0000
GPS-02	3° 29' 03,95340" N	76° 11' 52,87417" W	1.535,5173	-0,0009	0,0005	-0,0227
gps1	3° 29' 05,31531" N	76° 11' 53,04965" W	1.536,0853	0,0003	-0,0007	0,0045

Standard Deviations

Station	SD WGS84 Lat [m]	SD WGS84 Long [m]	SD Height [m]
BUEN	0,0124	0,0139	0,0000
CALI	0,0000	0,0000	0,0000
GPS-02	0,0078	0,0086	0,0189
gps1	0,0073	0,0098	0,0218

Absolute Error Ellipses (Confidence Levels: 2D - 39.4% 1D - 68.3%)

Station	Semi major (a) [m]	Semi minor (b) [m]	SD Height [m]	Orientation (φ) [°]
BUEN	-	-	0,0000	-
CALI	-	-	0,0000	-
GPS-02	-	-	0,0189	-
gps1	-	-	0,0218	-

Relative Error Ellipses (Confidence Levels: 2D - 39.4% 1D - 68.3%)

From	To	Semi major (a) [m]	Semi minor (b) [m]	SD Height [m]	Orientation (ψ) [°]
BUEN	GPS-02	0,0115	0,0102	0,0189	-9
BUEN	gps1	0,0164	0,0138	0,0218	-19
CALI	GPS-02	0,0087	0,0077	0,0189	30
CALI	gps1	0,0098	0,0073	0,0218	22

External Reliability

Station	Component	Ext. Rel. [m]	Observation Type	Station	Target
BUEN	EW	-0,1109	DX of Baseline	BUEN	GPS-02
	NS	-0,0848	DZ of Baseline	BUEN	GPS-02
	Height	0,0000	DY of Baseline	BUEN	GPS-02
CALI	EW	0,0000	DY of Baseline	CALI	GPS-02
	NS	0,0000	DY of Baseline	CALI	GPS-02
	Height	0,0000	DY of Baseline	BUEN	GPS-02
GPS-02	EW	0,1192	DX of Baseline	CALI	GPS-02
	NS	0,0942	DZ of Baseline	CALI	GPS-02
	Height	-0,0704	DY of Baseline	CALI	GPS-02
gps1	EW	0,1202	DX of Baseline	CALI	gps1
	NS	0,0951	DZ of Baseline	CALI	gps1
	Height	-0,2503	DY of Baseline	CALI	gps1

Adjusted GNSS Observations

From Station	Target Point	ΔX [m]	ΔY [m]	ΔZ [m]	Res. ΔX [m]	Res. ΔY [m]	Res. ΔZ [m]
BUEN	GPS-02	88.800,6673	16.728,8535	-43.785,3190	-0,0092	0,0415	-0,0039
CALI	GPS-02	36.084,5413	8.970,8537	12.024,7934	0,0049	-0,0221	0,0022
CALI	gps1	36.078,8086	8.971,4805	12.066,5934	-0,0004	0,0046	-0,0006
BUEN	gps1	88.794,9346	16.729,4803	-43.743,5190	0,0075	-0,1091	0,0123

Adjusted GNSS Observations - continued

From Station	Target Point	SD ΔX [m]	SD ΔY [m]	SD ΔZ [m]	W-Test ΔX	W-Test ΔY	W-Test ΔZ	T-Test
BUEN	GPS-02	0,0120	0,0184	0,0104	-0,04	1,68	-0,22	0,96
CALI	GPS-02	0,0096	0,0184	0,0079	0,04	-1,68	0,22	0,96
CALI	gps1	0,0113	0,0210	0,0075	0,29	1,07	-0,14	0,43
BUEN	gps1	0,0170	0,0213	0,0139	-0,29	-1,07	0,14	0,43

GNSS Baseline Vector Residuals

From Station	Target Point	Adj. Vector	Resid [m]	Resid [ppm]
BUEN	GPS-02	100.411,9874	0,0427	0,4
CALI	GPS-02	39.078,9712	0,0227	0,6
CALI	gps1	39.086,7058	0,0046	0,1
BUEN	gps1	100.388,8011	0,1100	1,1

Testing and Estimated Errors

Coordinate Tests

Station		MDB	Red	BNR	W-Test	Est Error	T-Test
BUEN	WGS84 Latitude	-	-	-	-	-	-
	WGS84 Longitude	-	-	-	-	-	
	Height	-	-	-	-	-	
CALI	WGS84 Latitude	-	-	-	-	-	-
	WGS84 Longitude	-	-	-	-	-	
	Height	-	-	-	-	-	

Observation Tests

	Station	Target	MDB	Red	BNR	W-Test	Est Error	T-Test
ΔX	BUEN	GPS-02	0,1275 m	10	10,19	-0,04	-	0,96
ΔY			0,1113 m	62	3,13	1,68	-	-
ΔZ			0,0995 m	10	8,71	-0,22	-	-
ΔX	CALI	GPS-02	0,1275 m	5	13,83	0,04	-	0,96
ΔY			0,1113 m	33	5,01	-1,68	-	-
ΔZ			0,0995 m	5	12,19	0,22	-	-
ΔX	CALI	gps1	0,1302 m	5	12,26	0,29	-	0,43
ΔY			0,2690 m	4	14,00	1,07	-	-
ΔZ			0,0997 m	4	13,02	-0,14	-	-
ΔX	BUEN	gps1	0,1302 m	85	1,20	-0,29	-	0,43
ΔY			0,2690 m	96	0,74	-1,07	-	-
ΔZ			0,0997 m	81	1,36	0,14	-	-

Antenna Height Tests

Station	Target	Date/Time	Antenna MDB	Antenna W-Test	Est. Antenna Error
BUEN	GPS-02	13/09/2026 09:45:00	0,1110 m	-1,65	-
CALI	GPS-02	13/09/2026 09:45:00	0,1110 m	1,65	-
CALI	gps1	04/10/2026 11:00:27	0,2978 m	-1,02	-
BUEN	gps1	04/10/2026 11:00:27	0,2987 m	1,03	-